SnapCenter Service Snapshot backup configuration

NetApp Solutions SAP

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SnapCenter Service Snapshot backup configuration

SnapCenter Service Snapshot backup configuration overview

Previous: SnapCenter Service policy configuration.

SnapCenter Service supports two different Snapshot backup operation types:

- SAP HANA data backups by using Snapshot operations.

  Backup of the SAP HANA database data volume combining HANA SQL statements for database consistency and backup catalog management with Azure NetApp Files Snapshot operations.

- Backup of volumes with unstructured data by using Snapshot backup operations.

  Backup of binaries, configuration files, and so on, which do not require any interaction with the HANA database. The backup is executed with an Azure NetApp Files Snapshot backup operation. In SnapCenter Service, two sub-types are used to define whether these volumes belong to a specific HANA system or whether it is a global shared volume used by multiple HANA systems.

  - Backup of non-data volumes of an SAP HANA database

    Backup of a volume that is associated with a specific HANA system, for example, the HANA shared volume.

  - Backup of global non-data volumes

    Backup of a volume that is not associated with a specific HANA system, for example, the SAP transport directory.

    In the following sections, the configuration of the three different backup types is described.

Next: Backup configuration of SAP HANA database backups.

Backup configuration of SAP HANA database backups

Previous: SnapCenter Service Snapshot backup configuration overview.

This section describes the configuration steps for a single host SAP HANA MDC single-tenant system PR1. The steps are identical for an MDC system with multiple tenants. The differences for a single-container or a multiple-host system are reflected in the corresponding configuration steps.

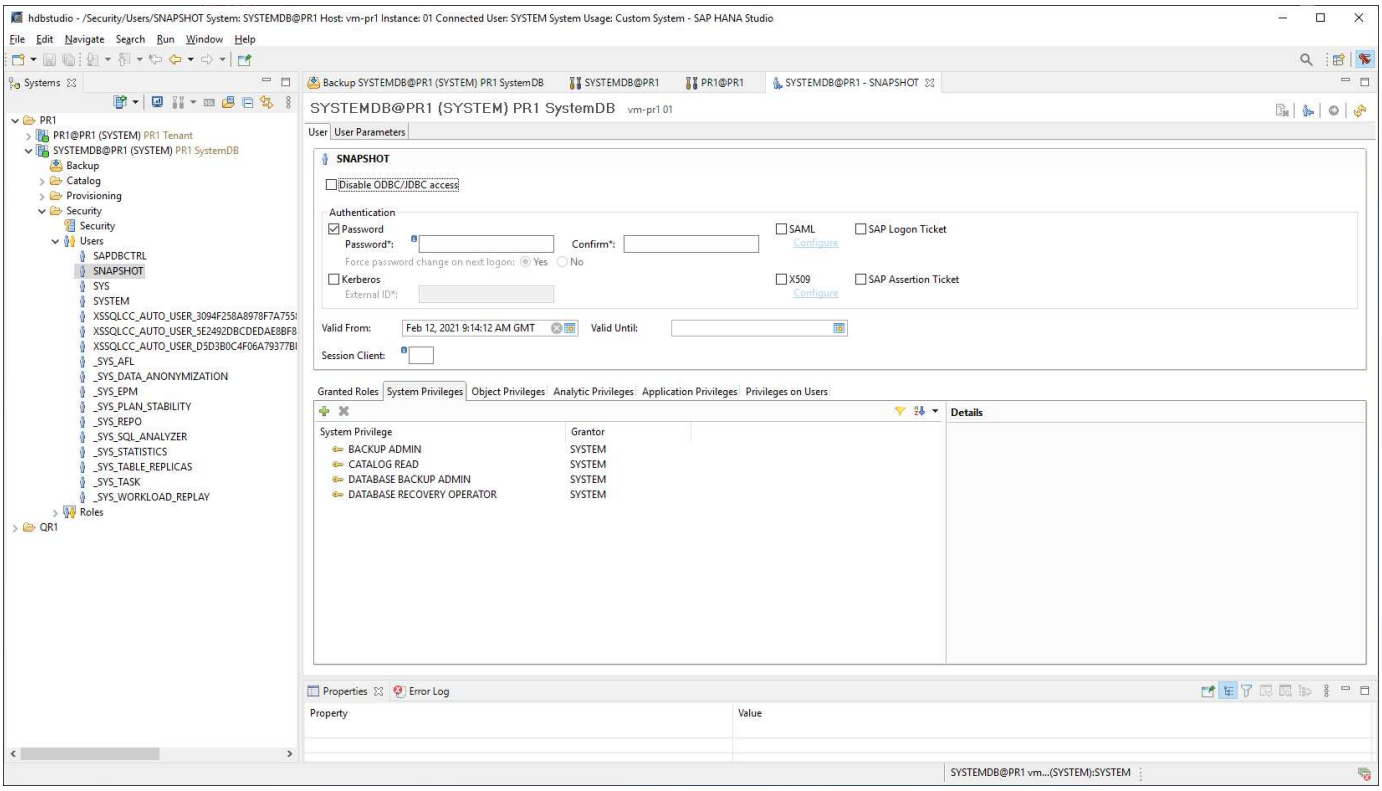
SAP HANA backup user

NetApp recommends configuring a dedicated database user in the HANA database to run the backup operations with SnapCenter Service. During the HANA system configuration in SnapCenter Service, an HANA database user store key is configured for this backup user, and this user store key is used to communicate with the HANA system.

The following figure shows a screenshot of SAP HANA Studio through which you can create the backup user.

> The required privileges changed in the SAP HANA 2.0 SPS5 release: backup admin, catalog read, database backup admin, and database recovery operator is required. For earlier releases, backup admin and catalog read are sufficient.

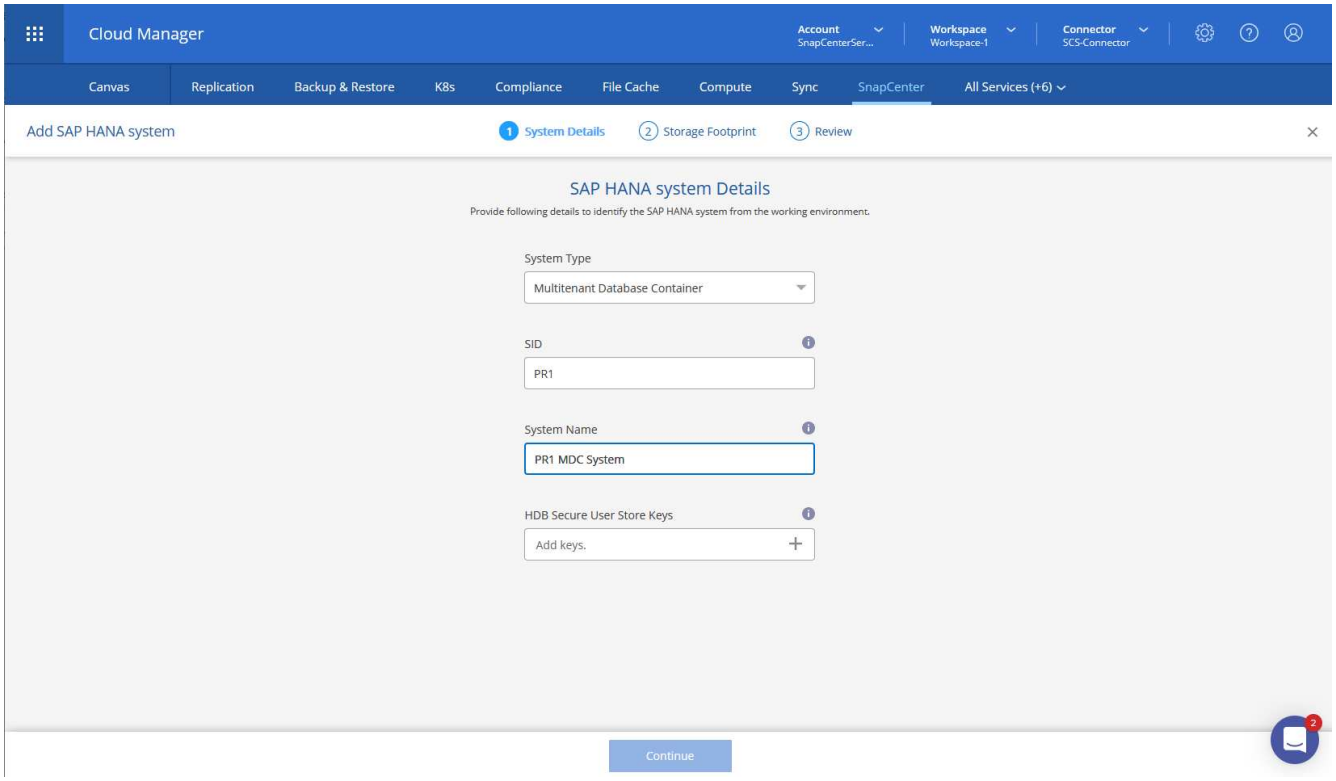
For an SAP HANA MDC system, with single or multiple tenants, you must create the user in the system database because all backup commands for the system and the tenant databases are executed by using the system database.



SAP HANA system configuration in SnapCenter Service

To configure the SAP HANA system PR1 in SnapCenter Service, follow these steps:

1. Go to SAP HANA Systems and click Add.
2. In the System Details screen, select the system type, HANA Single Container, or Multitenant Database Container. Provide the SID of the HANA system and enter any text of your choosing as a system name.
3. To add a HANA database user store key, Click Add Keys.

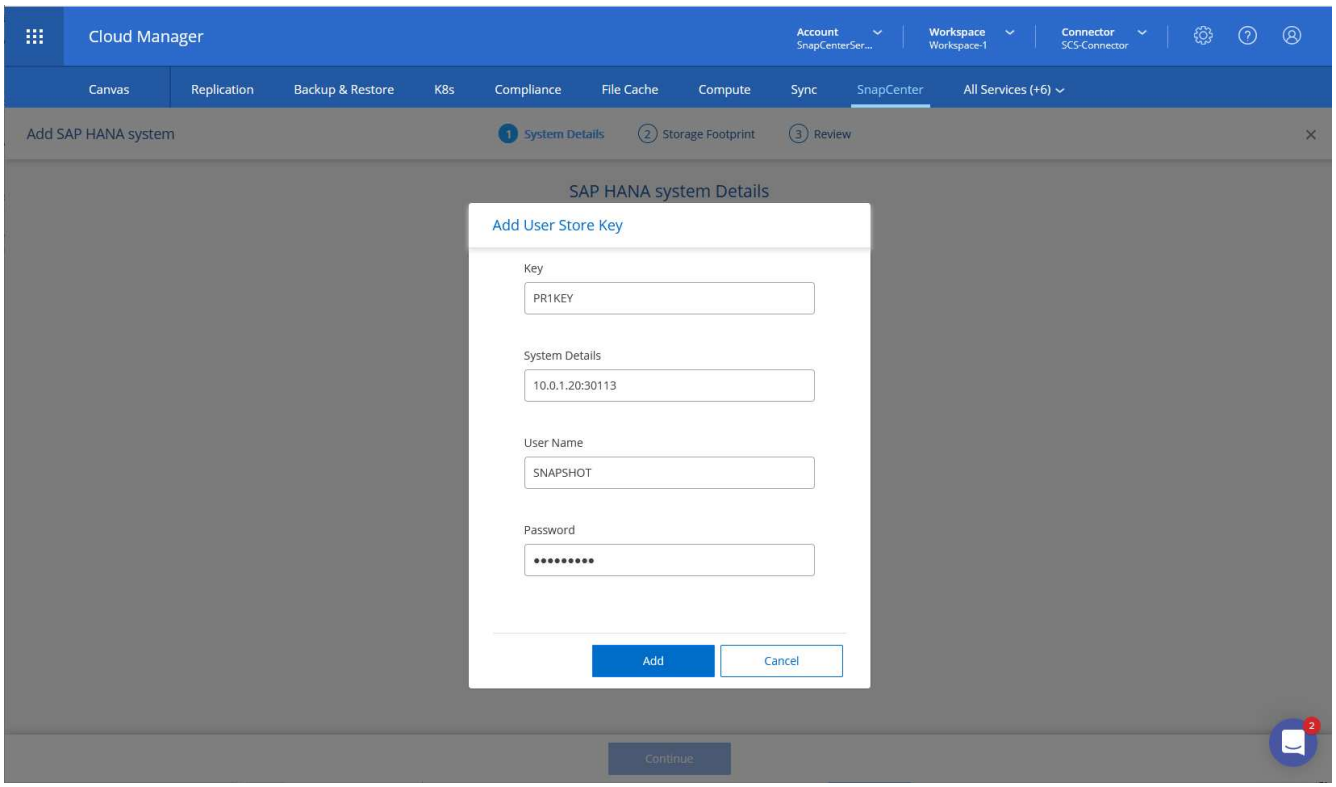


4. Enter the information for configuring the user store key for the HANA database. You can provide any name as the key name.

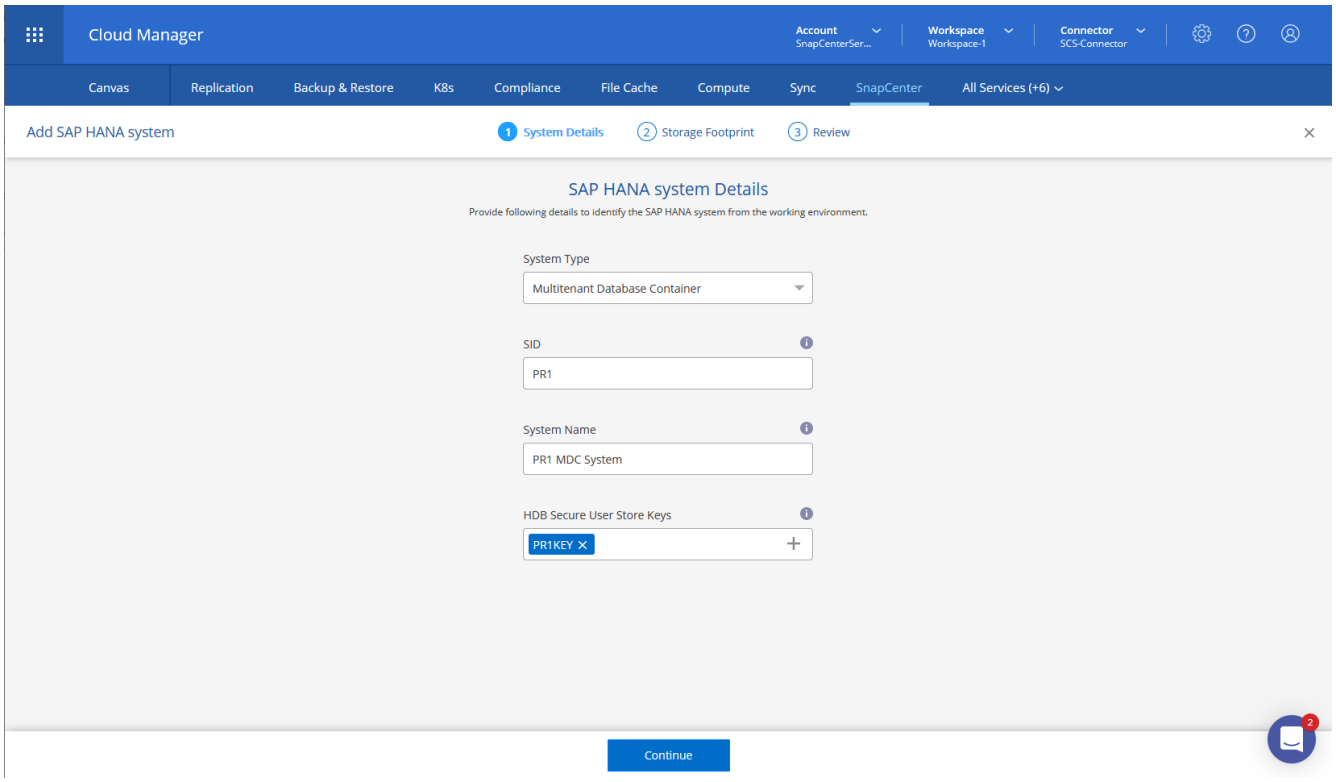
   The system details include the IP address and the port to communicate with the system database using the hdbsql client. In an SAP HANA MDC single-tenant setup, port `3<instanceNo>13` is the standard port for SQL access to the system database. For an SAP HANA single-container setup, port `3<instanceNo>15` is the standard port for SQL access to the index server and must be used in the configuration. For an SAP HANA multiple-host setup, you must configure user store keys for all hosts. SnapCenter Service tries to connect to the database by using each of the provided keys and can therefore operate independently of a failover of an SAP HANA service to a different host.

   You must provide the username and password for the database user in the system database that was previously configured. SnapCenter automatically creates a user store key by using this information and uses the key to communicate with the HANA database.

5. To continue, click Add.



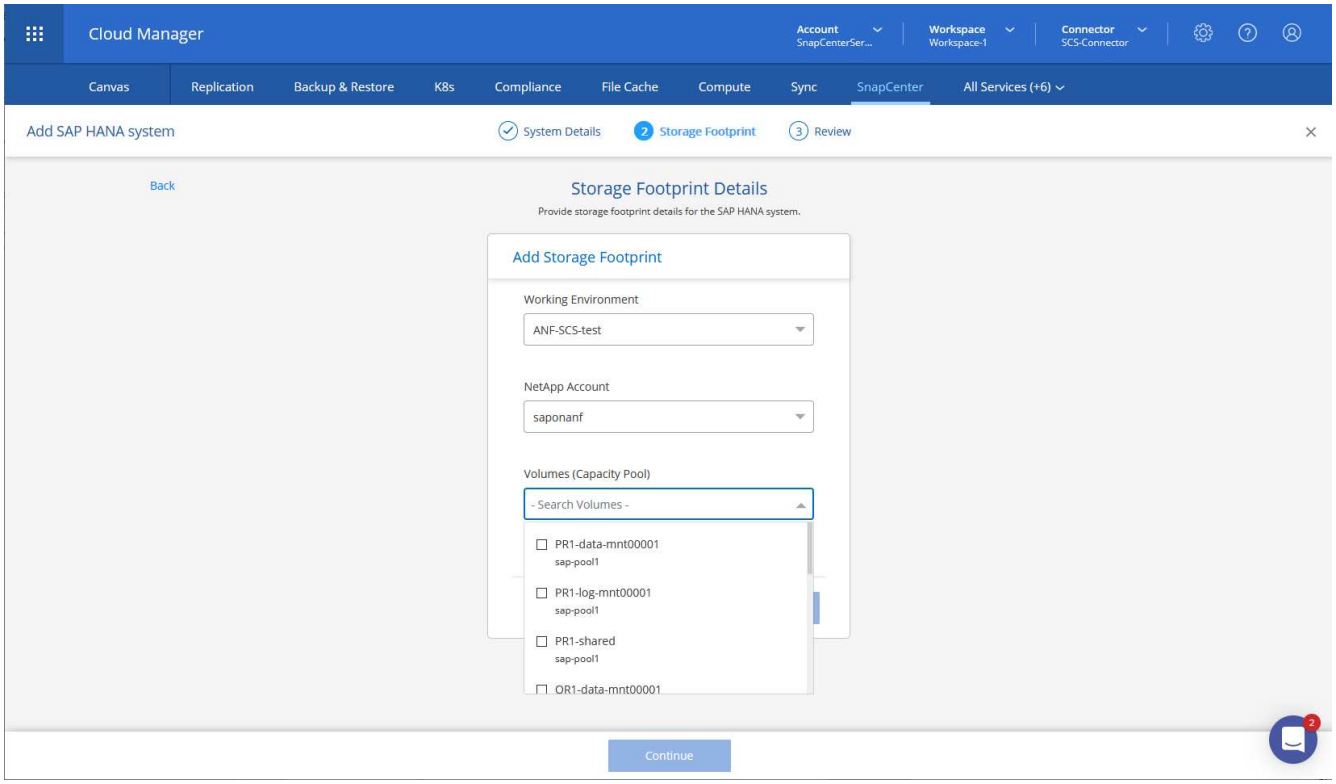
6. To proceed to the storage footprint configuration, click Continue.

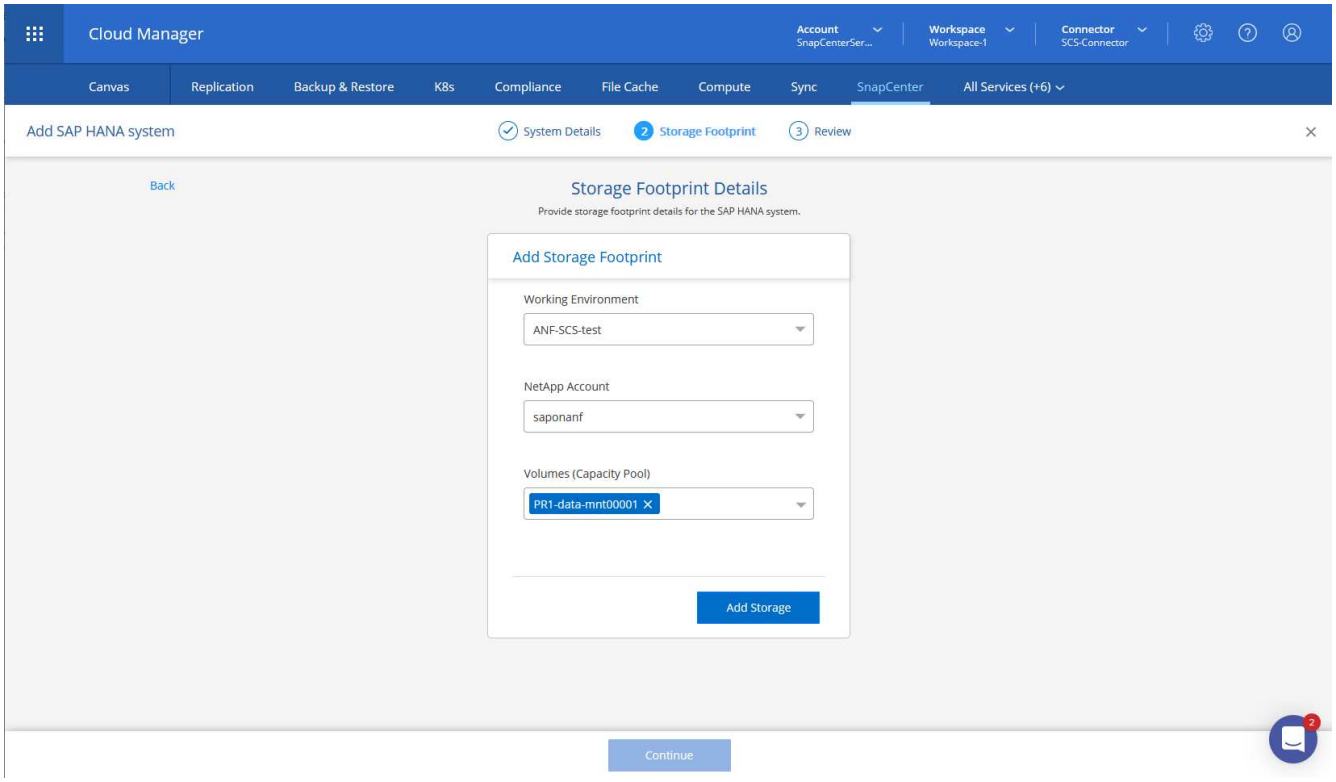


7. Enter the information for the storage volumes of the HANA system.

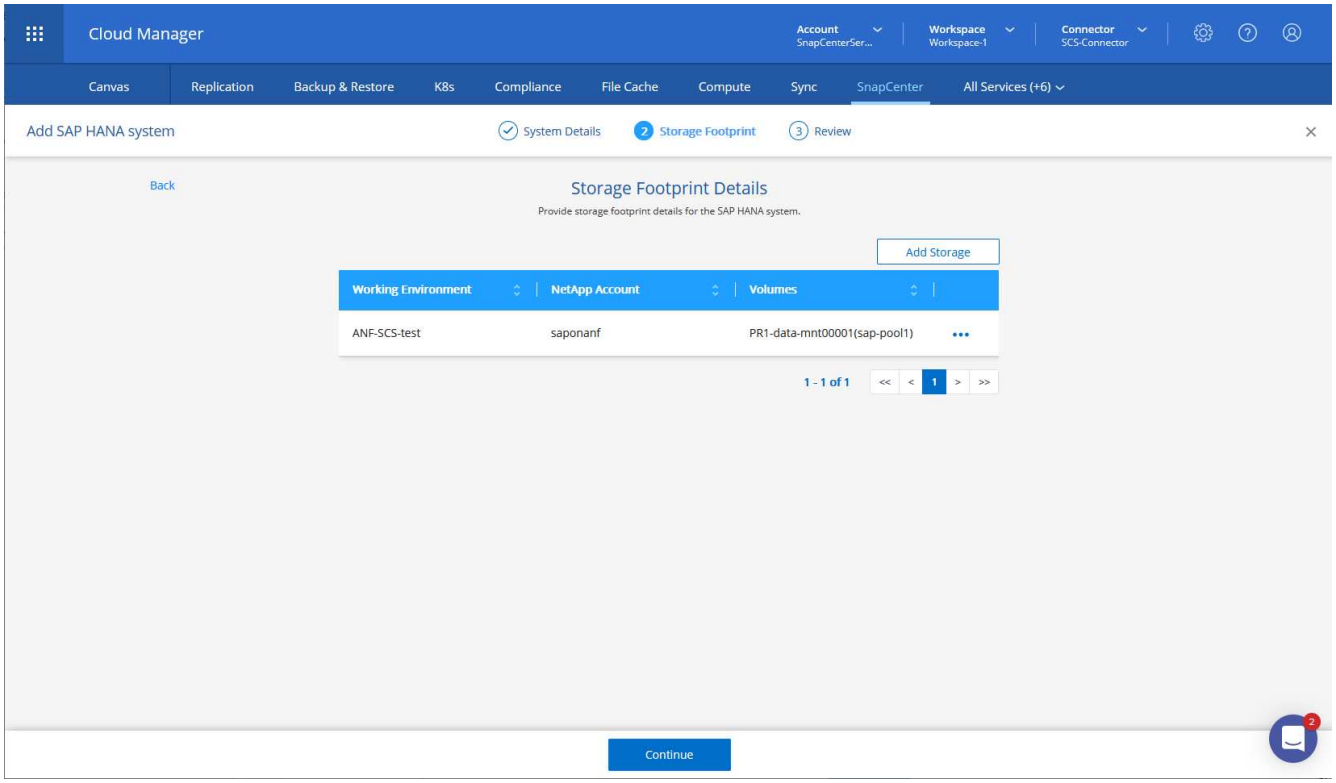
   Select the working environment and the NetApp account that is used for the Azure NetApp Files volumes of the HANA system. Select the data volume of your HANA system, for example, `PR1_data_mnt00001`.

For an SAP HANA multiple-host system, you must select the data volumes of all HANA hosts belonging to the system.

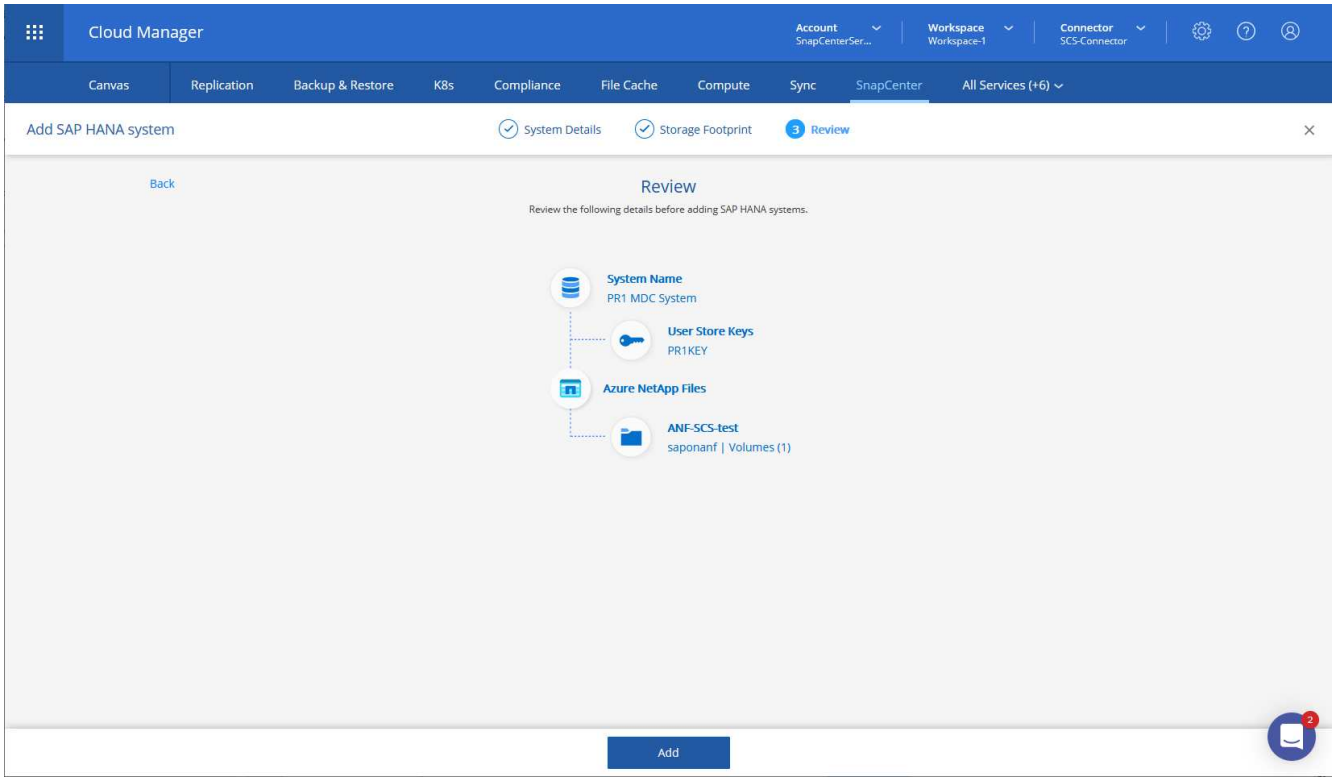




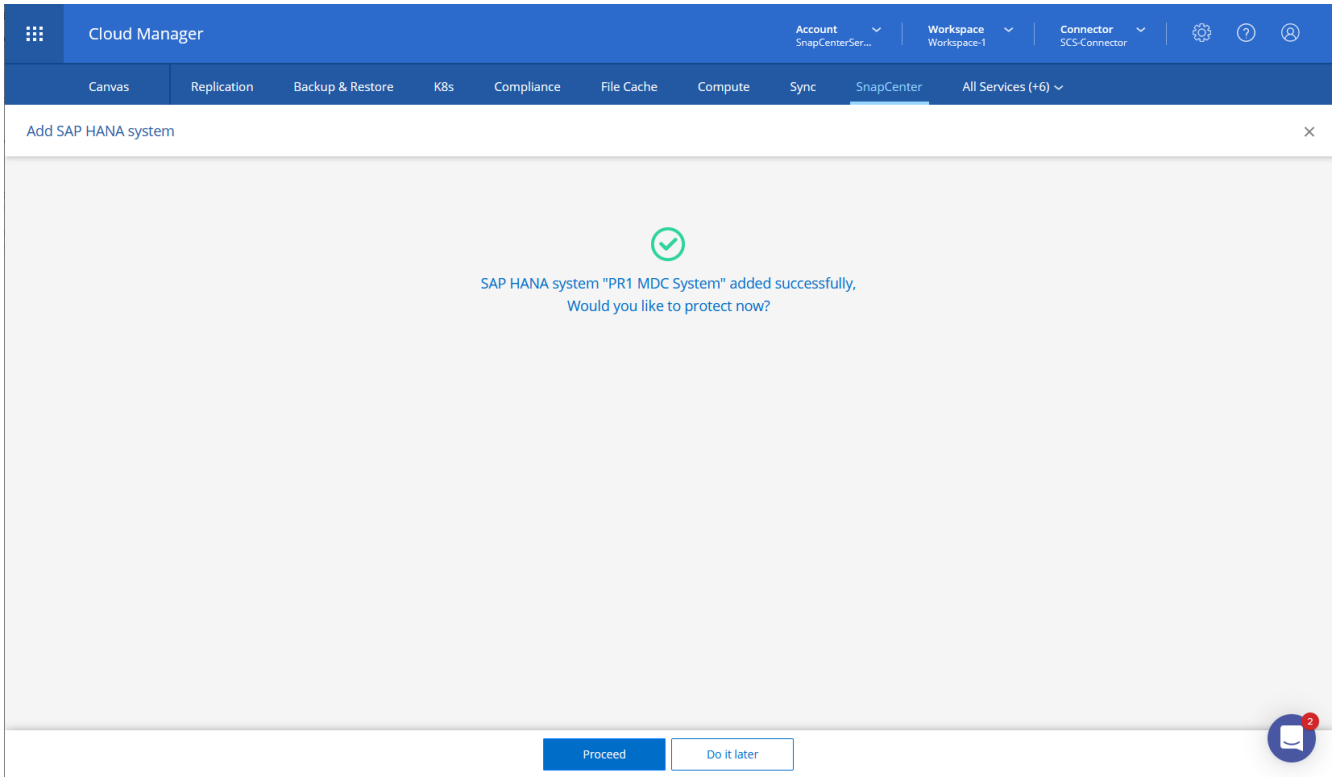
8. Click Continue.



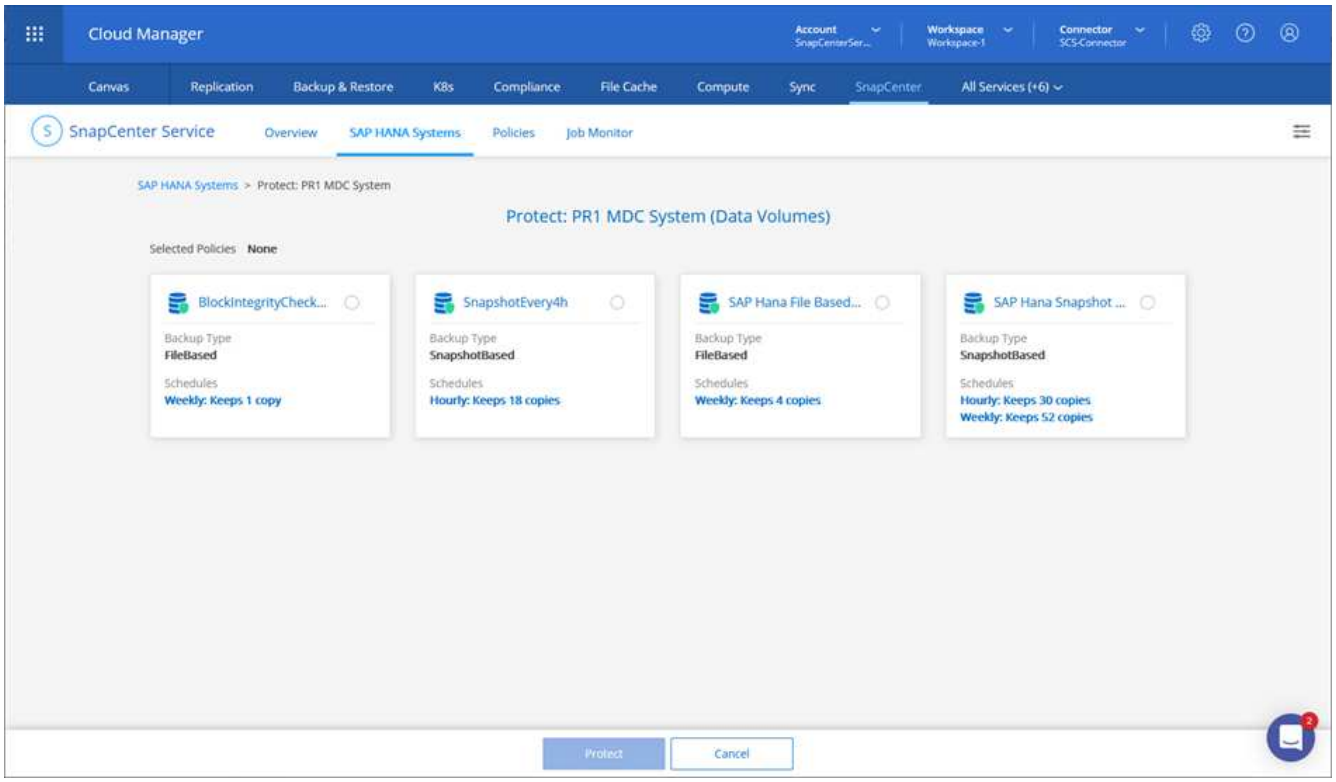
9. To add the HANA system, click Add.



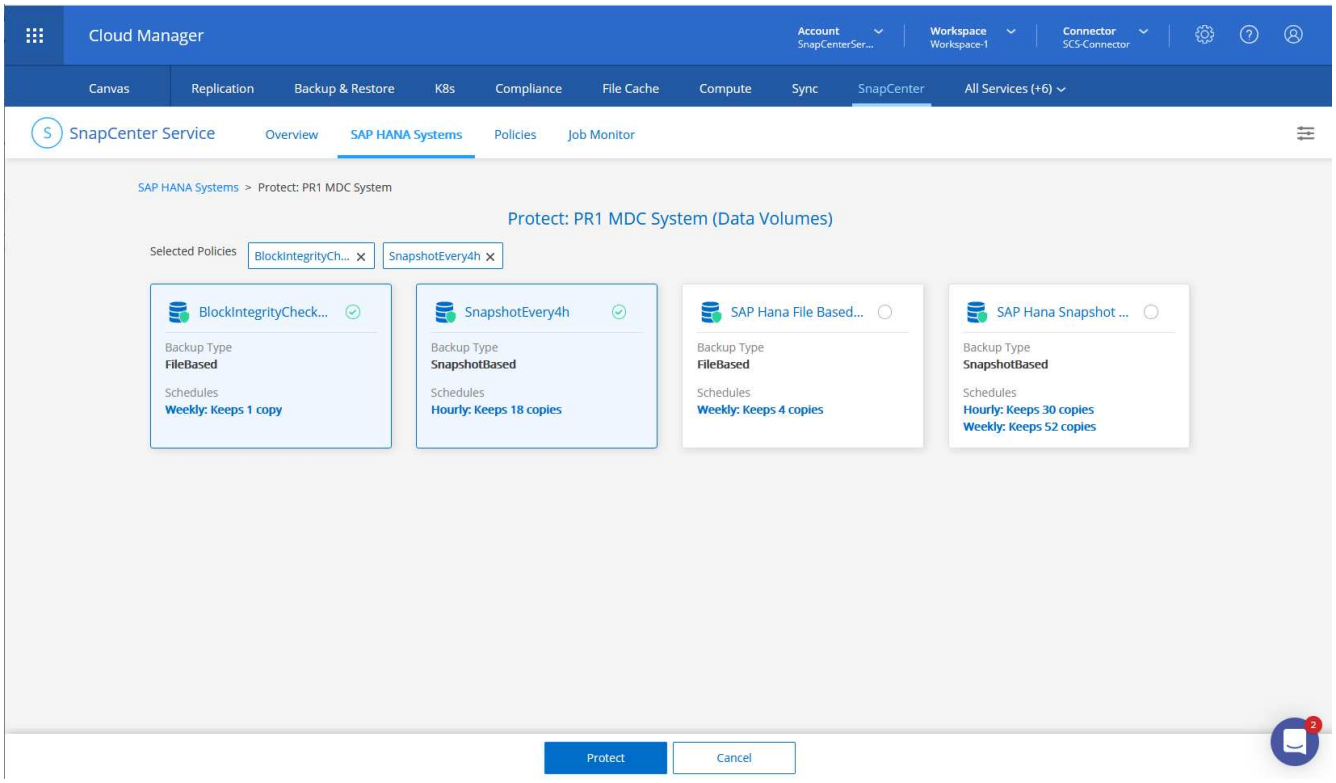
10. To continue with the protection configuration, click Proceed.



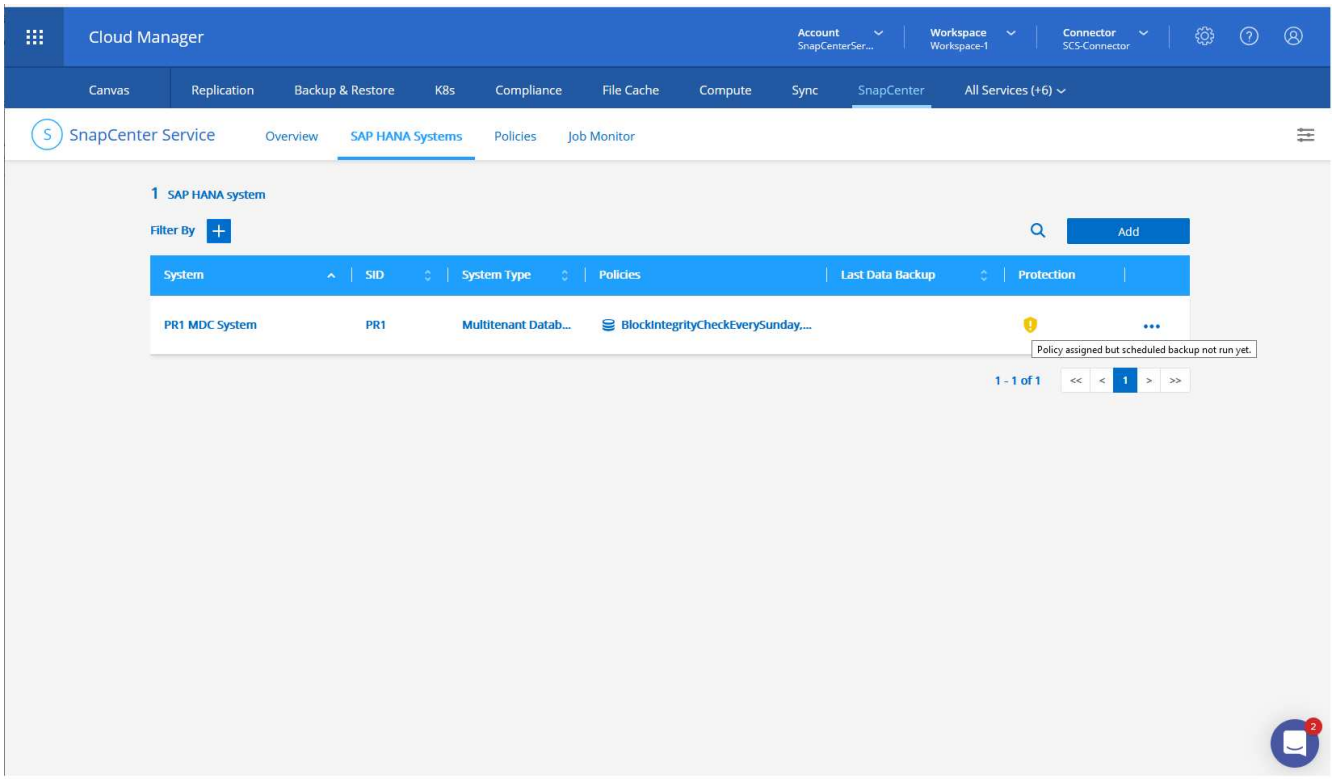
The screen displays all the policies that have been configured in SnapCenter Service.



11. Select the policies that you want to assign. For example, the policies `SnapshotEvery4h` and `BlockIntegrityCheckEverySunday` that were previously created are selected.
12. To assign the policies to the HANA system, click Protect.



The HANA systems overview screen shows the HANA system and the assigned policies.



Next: Backup configuration of SAP HANA non-data volumes.

Backup configuration of SAP HANA non-data volumes

Previous: Backup configuration of SAP HANA database backups.

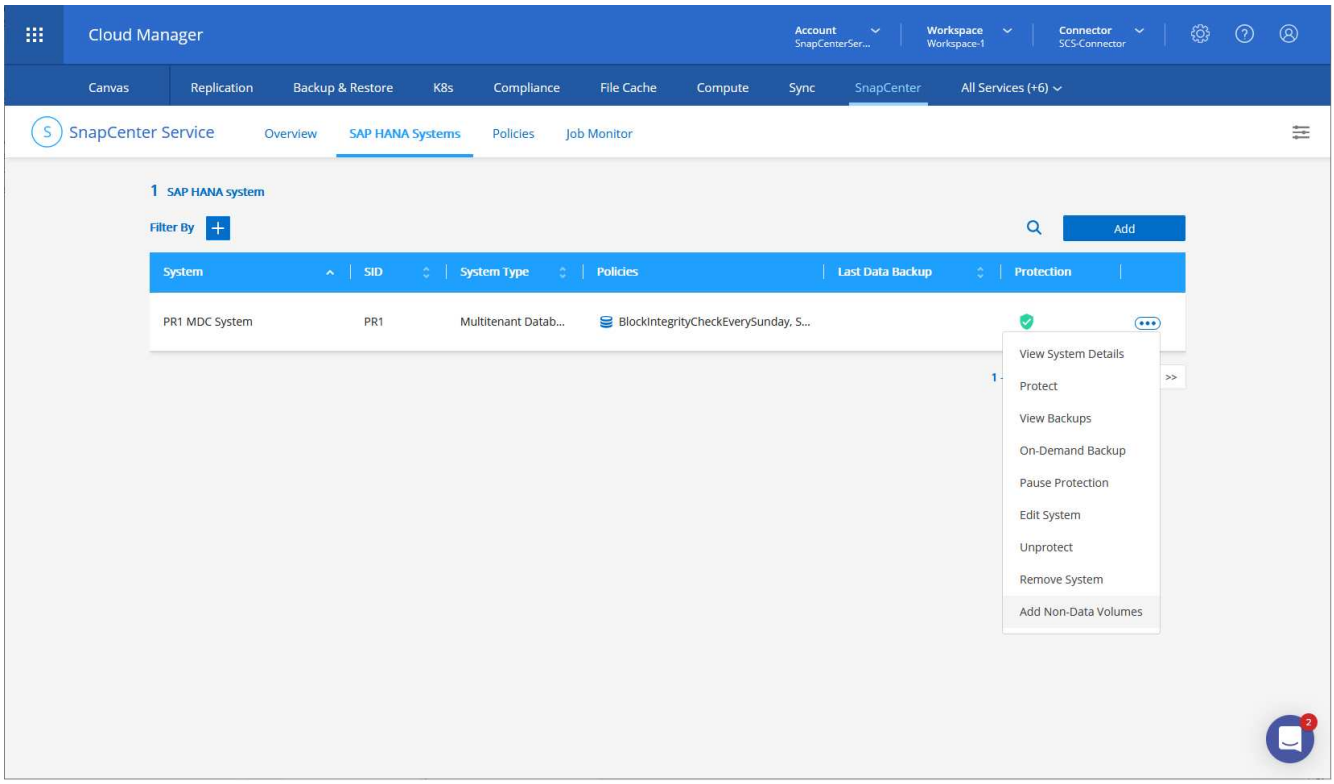
The backup of non-data volumes is an integrated part of the SnapCenter Service. Protecting the database data volume is sufficient to restore and recover the SAP HANA database to a given point in time provided that the database installation resources and the required logs are still available.

To recover from situations where other non-data files must be restored, NetApp recommends developing an additional backup strategy for these volumes to augment the SAP HANA database backup. Depending on your specific requirements, the backup of non-data volumes might differ in scheduling frequency and retention settings, and you should consider how frequently files are changed. For instance, the HANA volume `/hana/shared` contains executables but also SAP HANA trace files. Although executables only change when the SAP HANA database is upgraded, the SAP HANA trace files might need a higher backup frequency to support analyzing problem situations with SAP HANA.

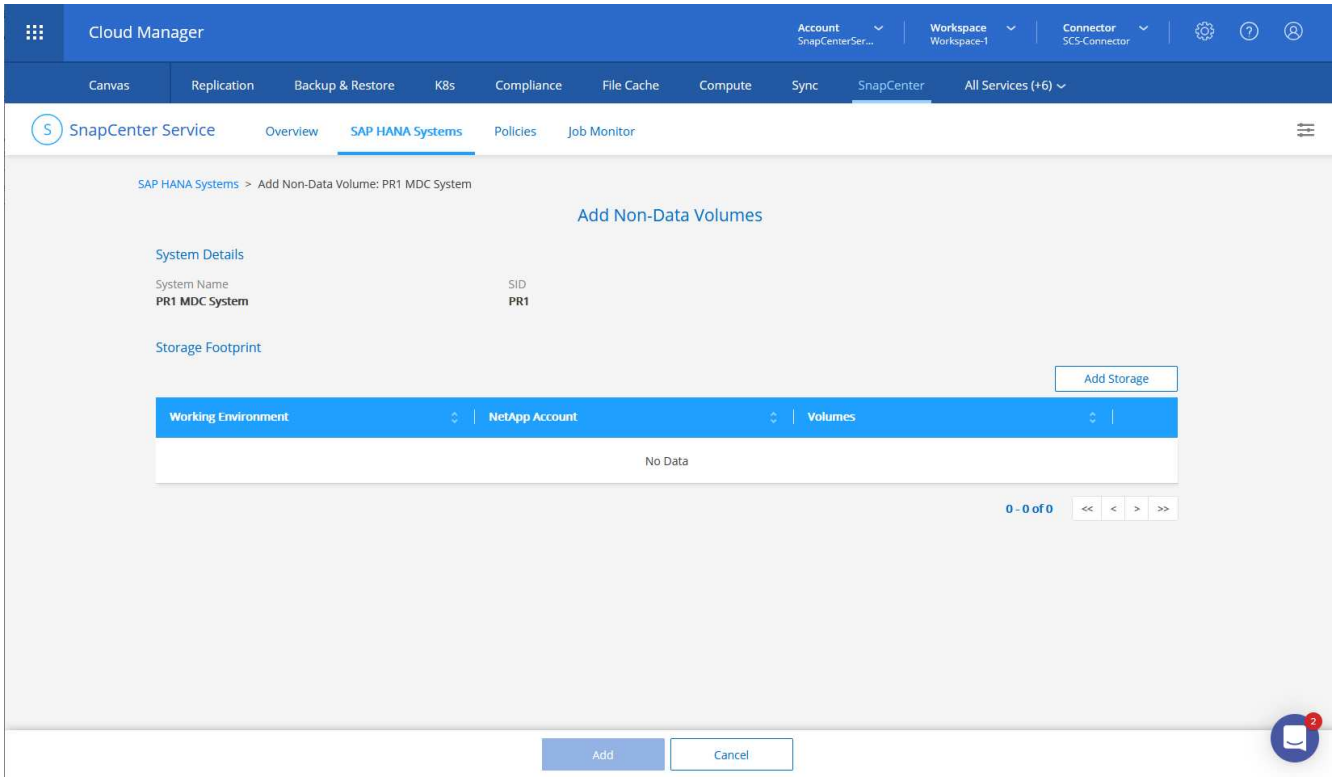
A SnapCenter Service non-data volume backup enables Snapshot copies of all relevant volumes to be created in a few seconds with the same space efficiency as SAP HANA database backups. The difference is that there is no SQL communication with the SAP HANA database required.

The following steps show how to protect the HANA shared volume of the SAP HANA database PR1.

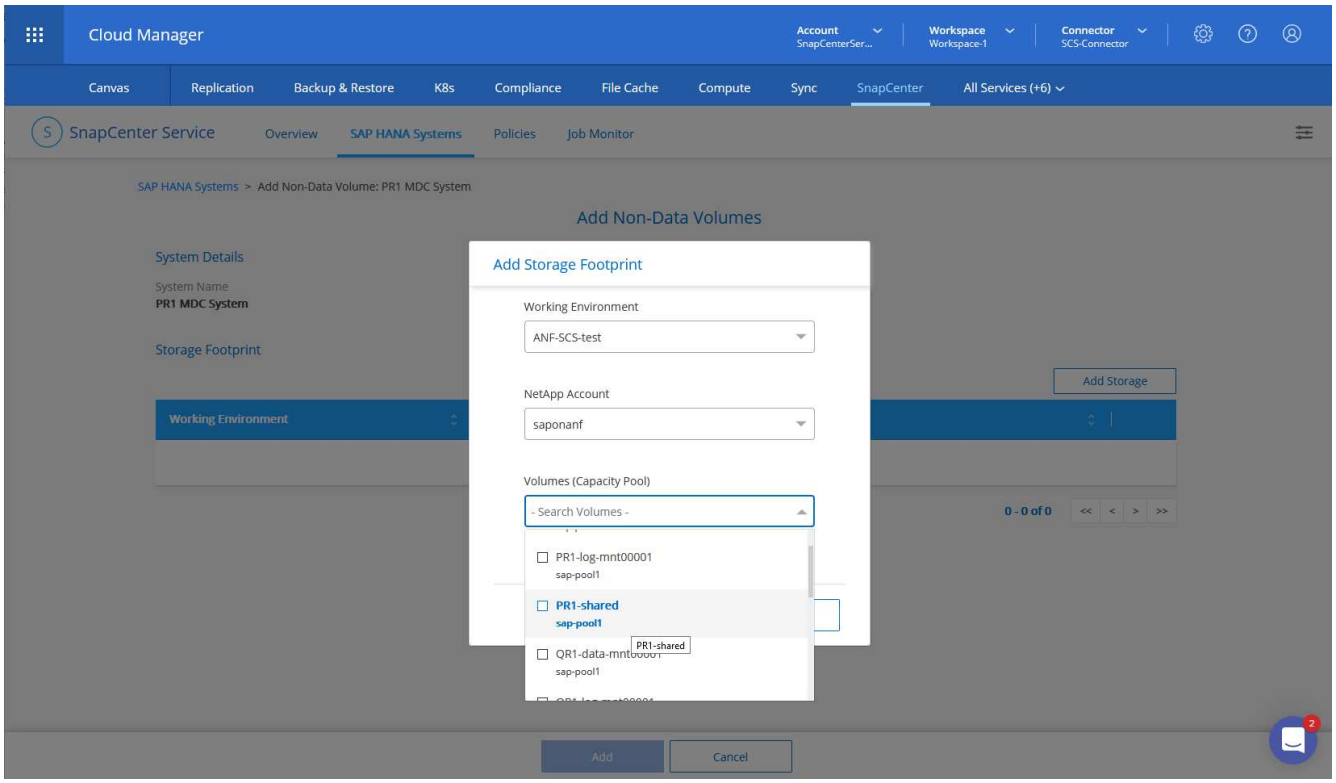
1. In the SAP HANA Systems view, select Add Non-Data Volumes in the context menu on the right of the screen.

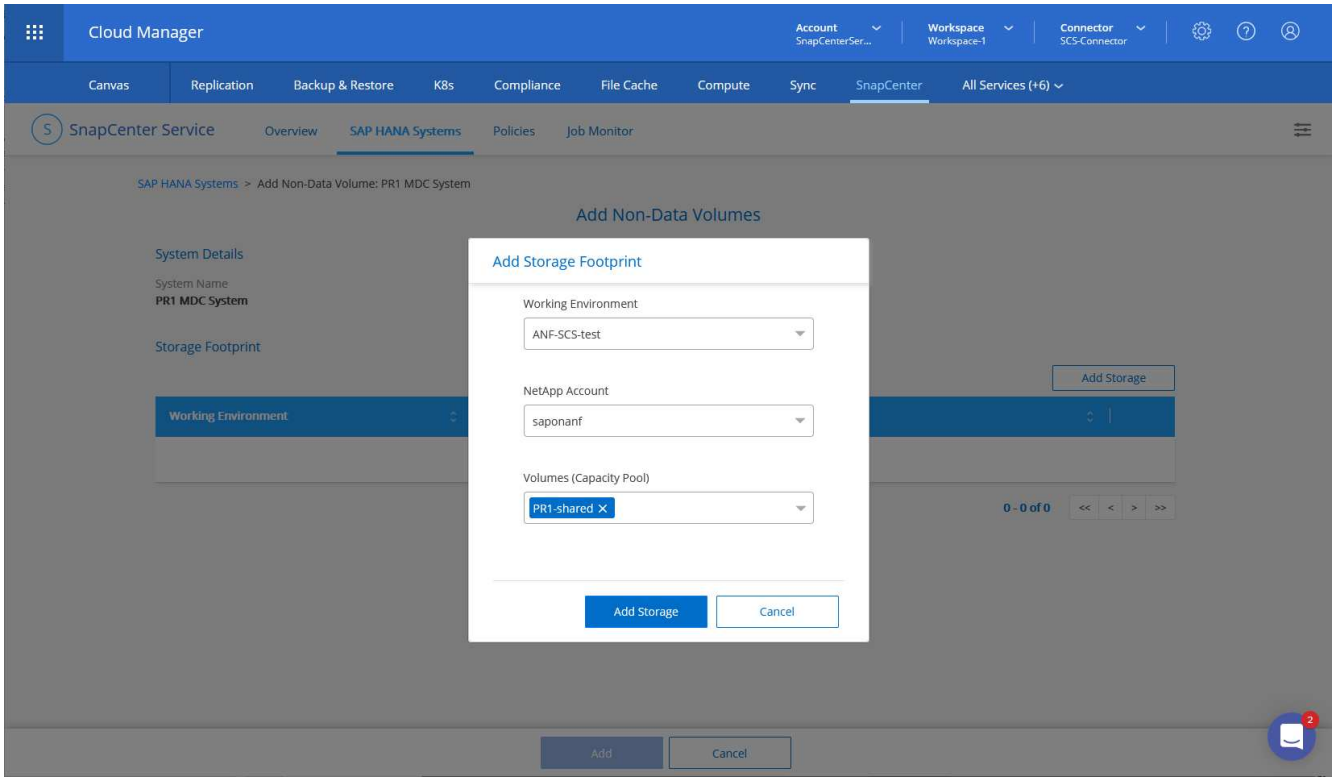


2. Click Add Storage.

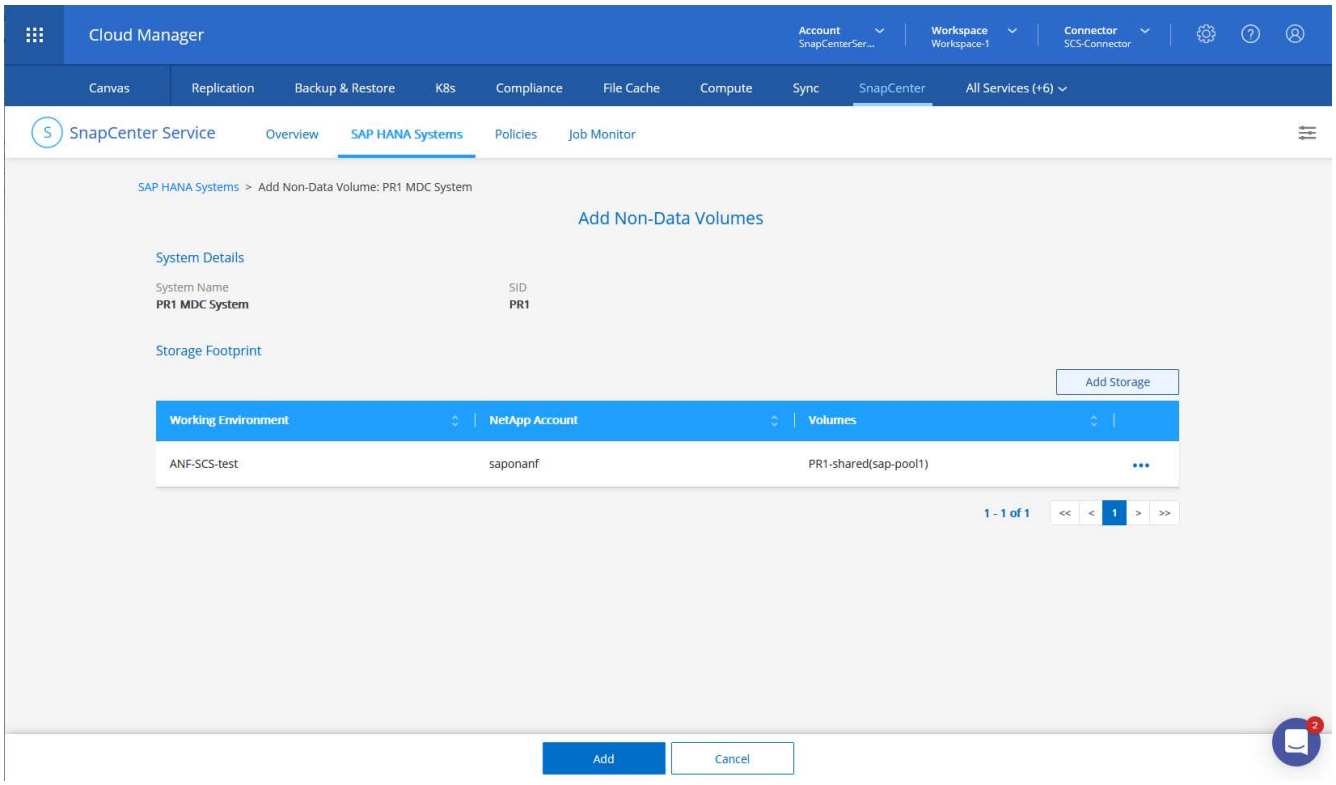


3. Select the HANA shared volume and click Add Storage.

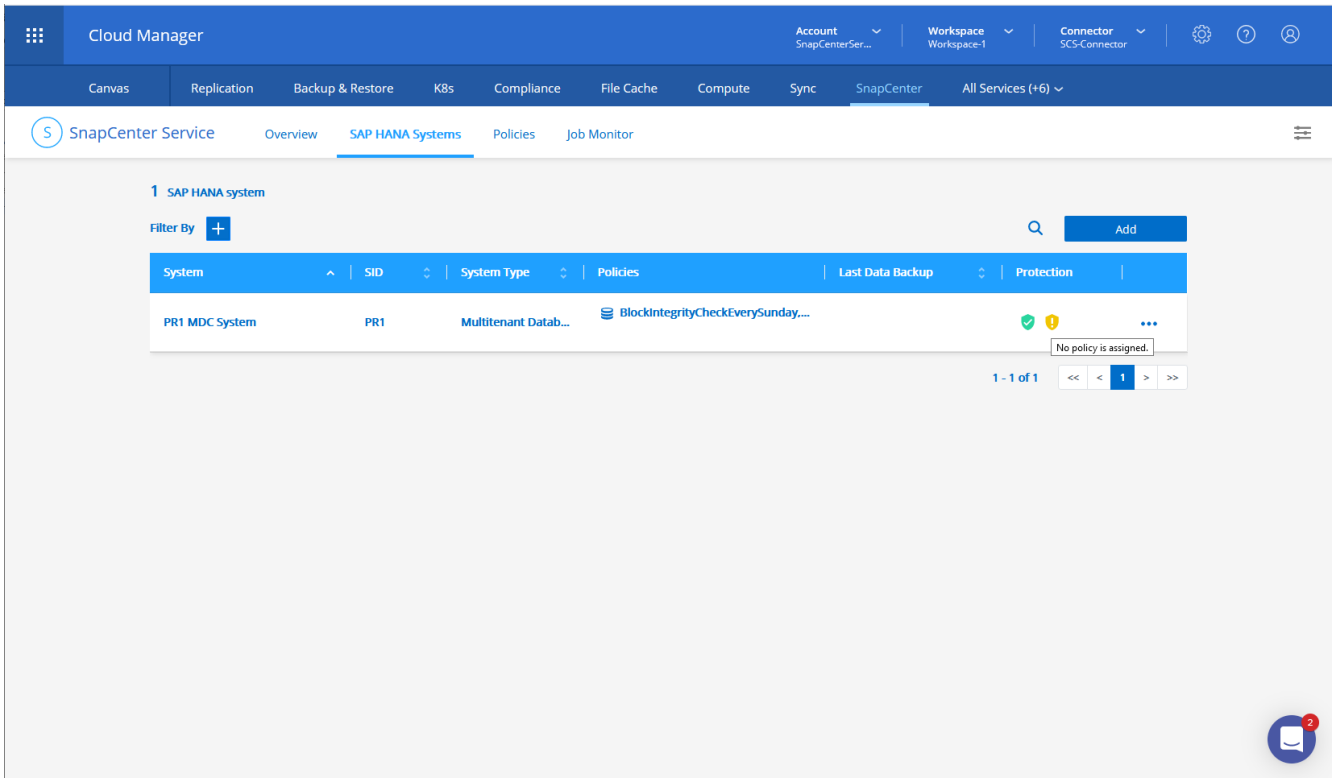




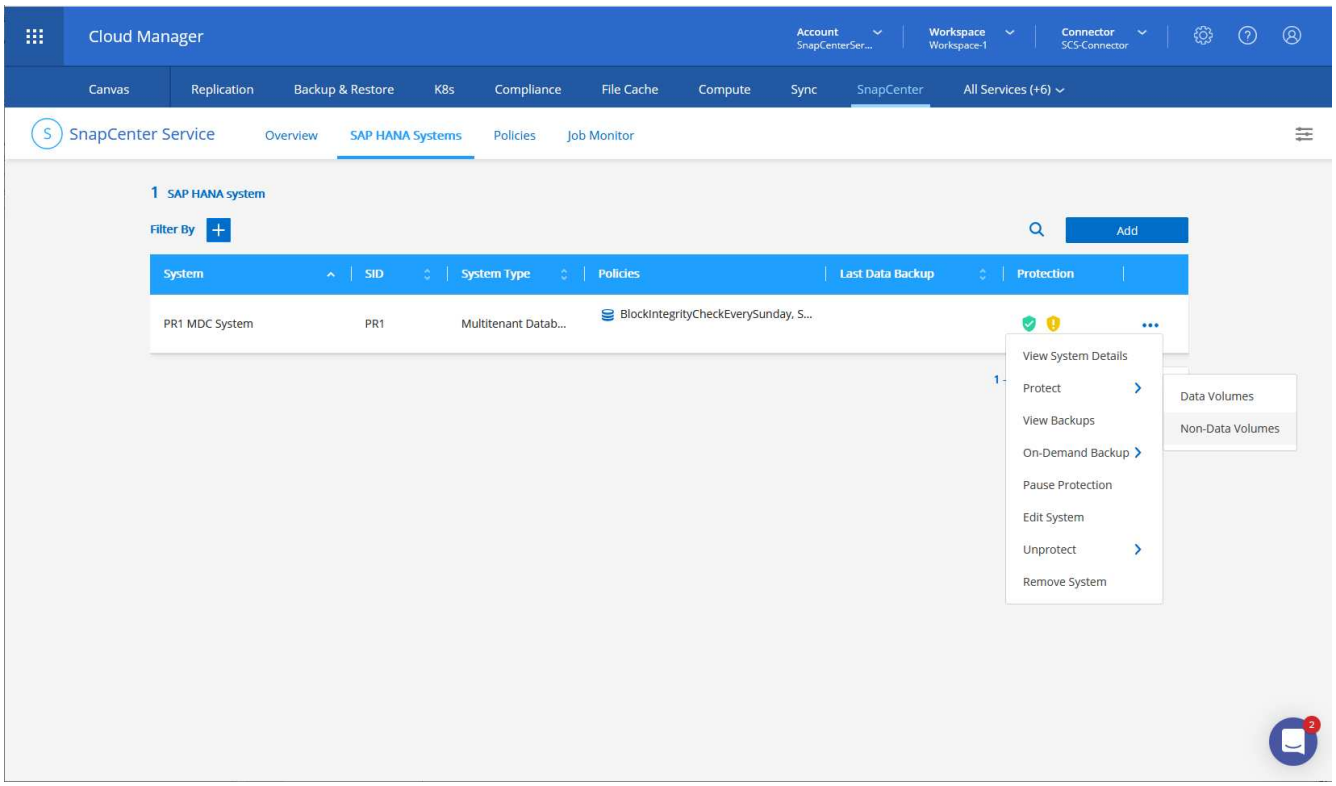
4. To add the non-data volume, click Add.



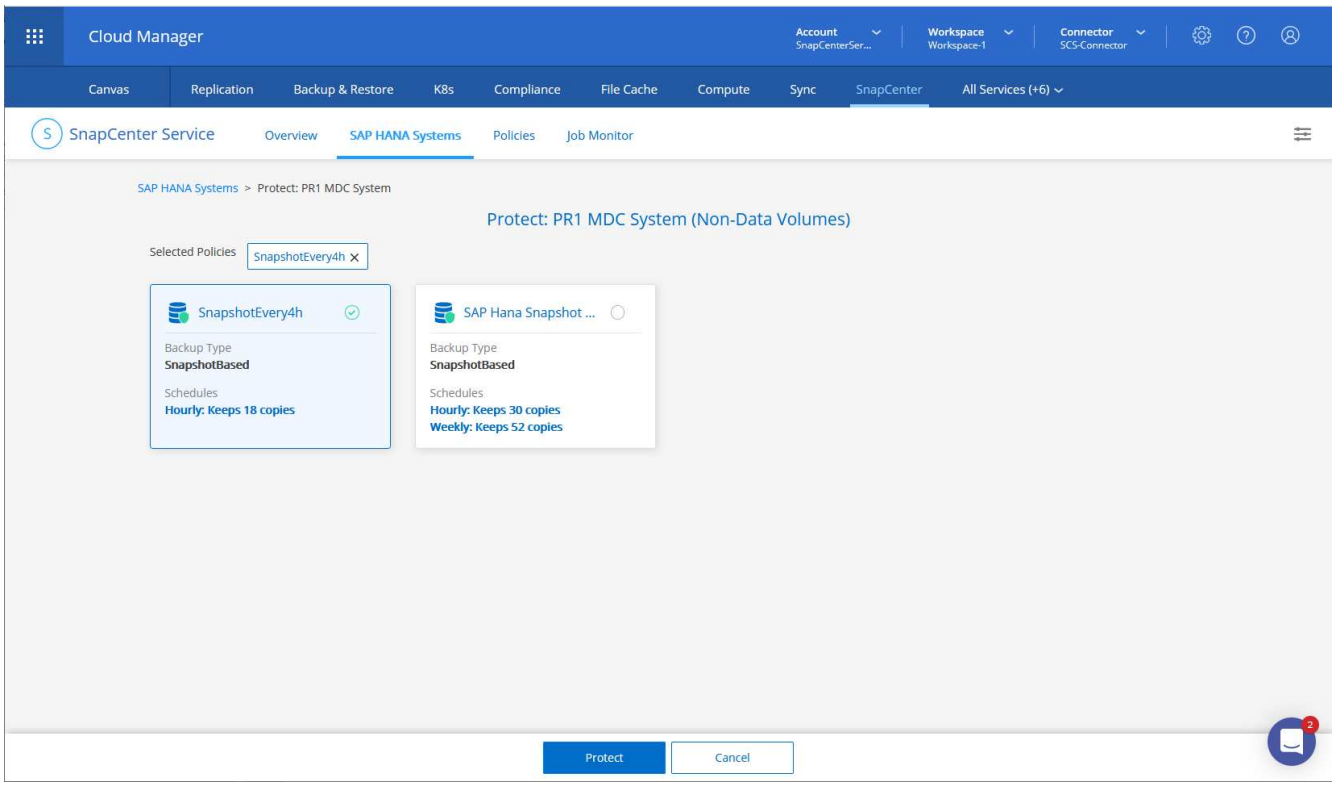
The SAP HANA Systems overview now shows a second icon in the protection column, which indicates that a non-data volume was added to the HANA system.



5. In the context menu on the right, select Protect > Non-Data Volumes.



6. Assign a policy to the non-data volume. In the following example, the `SnapshotEvery4h` policy is assigned to the HANA shared volume.



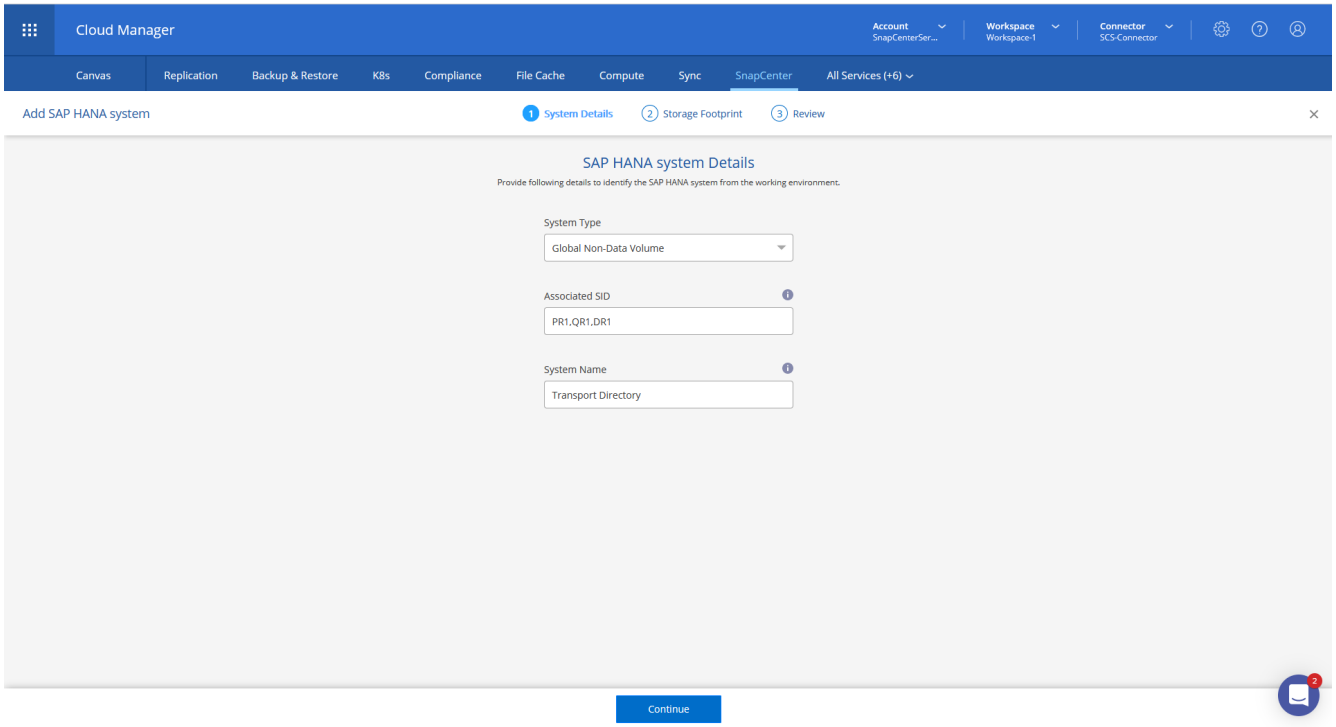
Next: Backup configuration of global non-data volumes.

Backup configuration of global non-data volumes

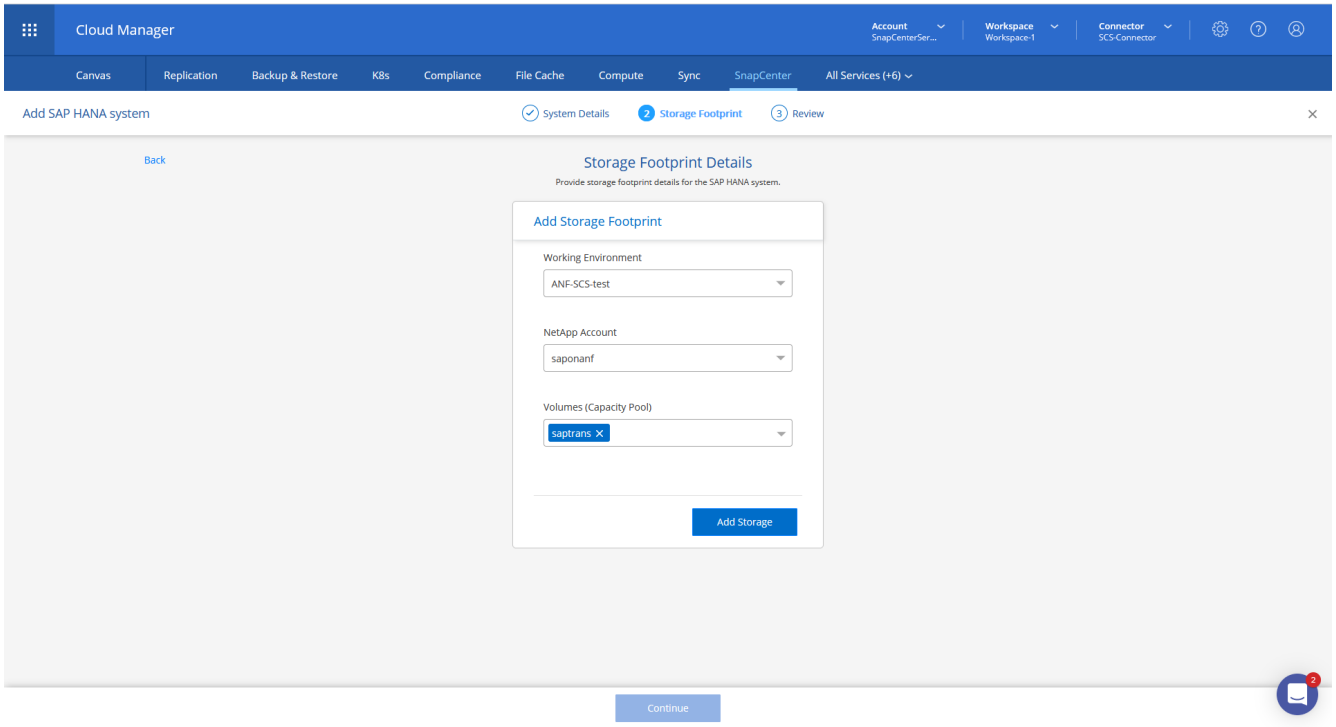
Previous: Backup configuration of SAP HANA non-data volumes.

To set the backup configuration of global non-data volumes, follow these steps:

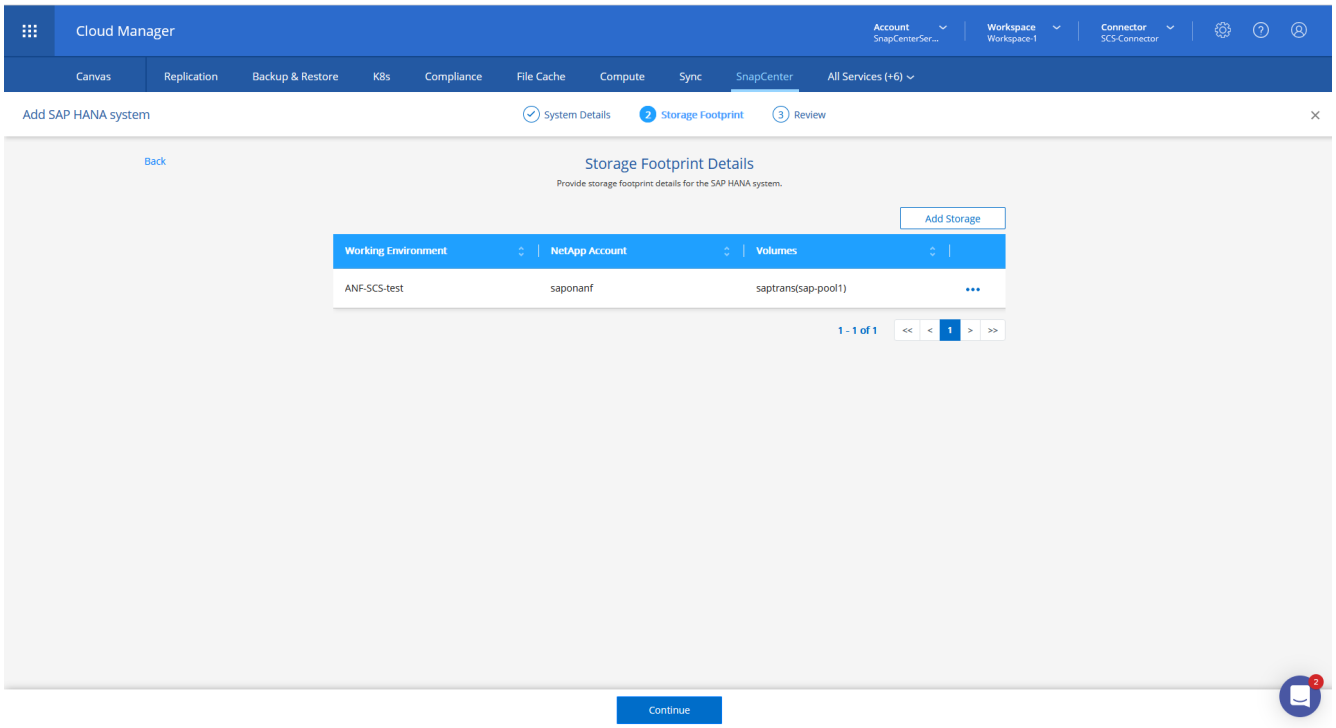
1. Go to SAP HANA Systems and click Add.
2. In the System Details screen, select the system type, Global Non-Data Volume. You can provide a comma-separated list of SIDs, which are associated with the global volume. Enter a system name of your choosing and click Continue.



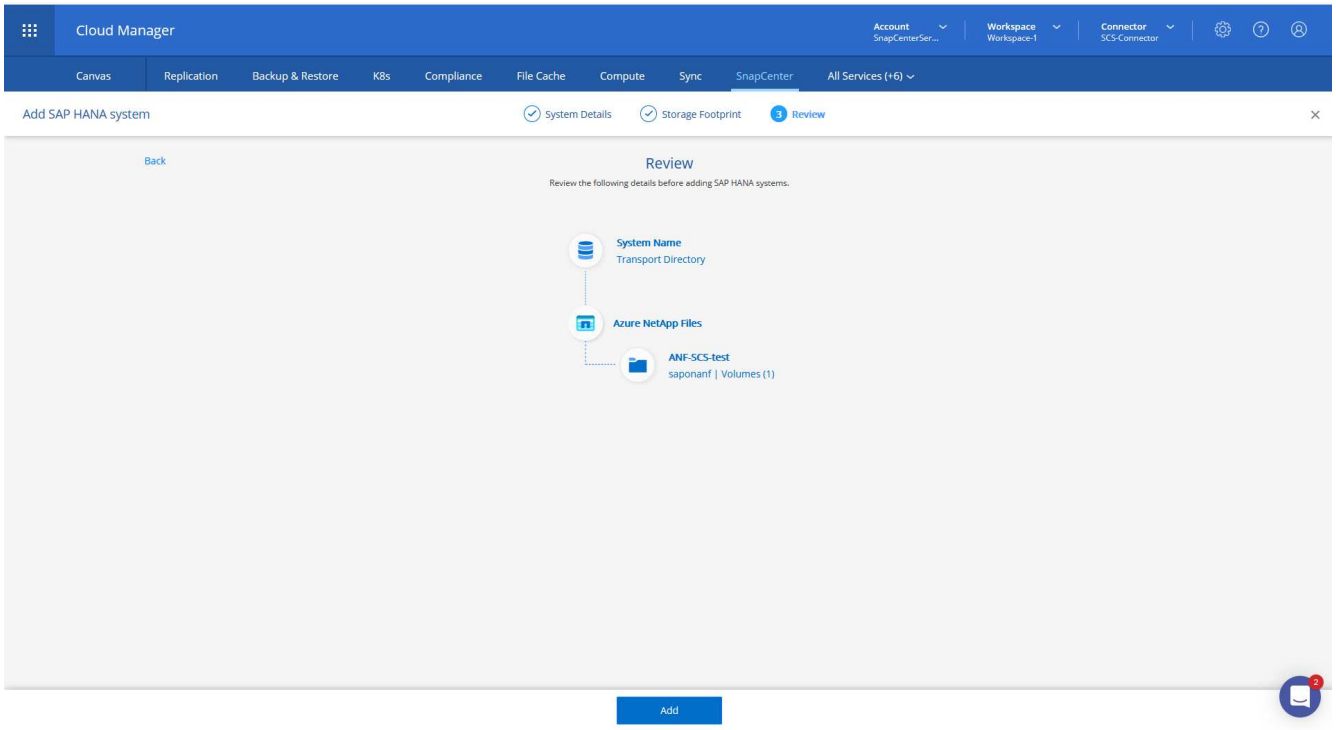
3. Select the volume and click Continue.



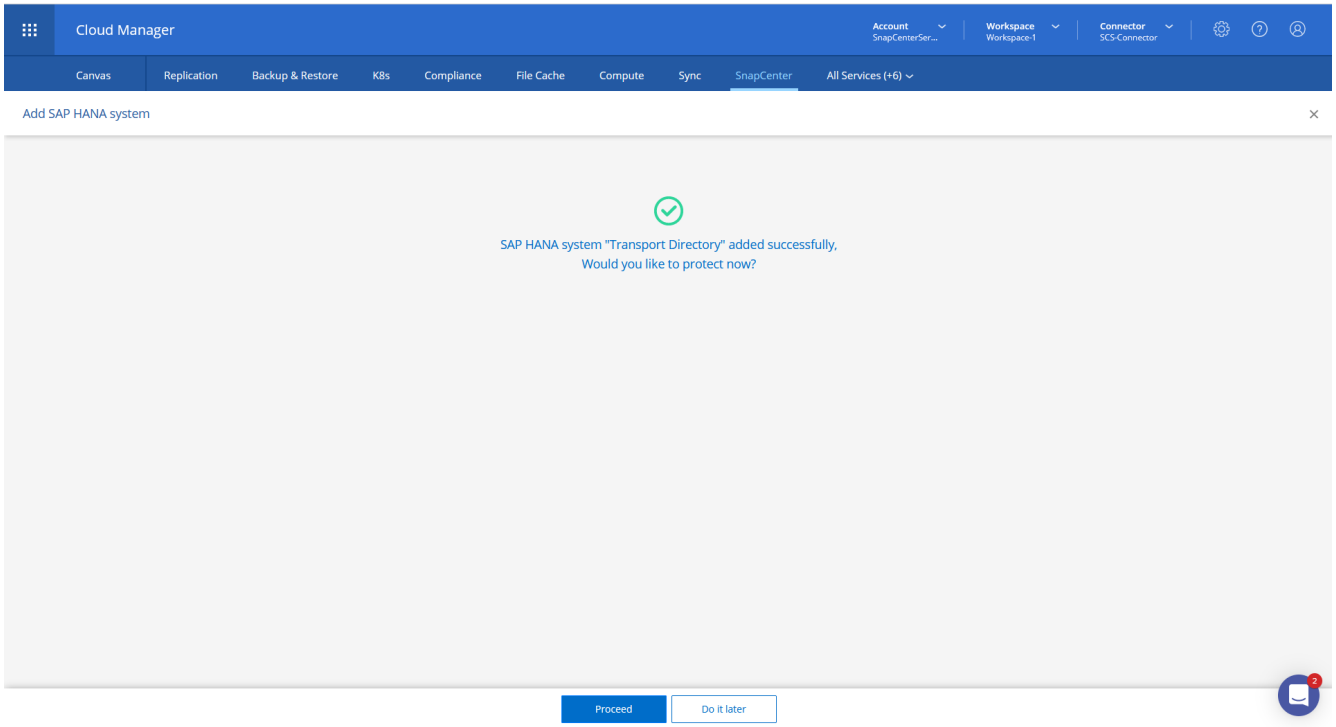
4. Click Continue.



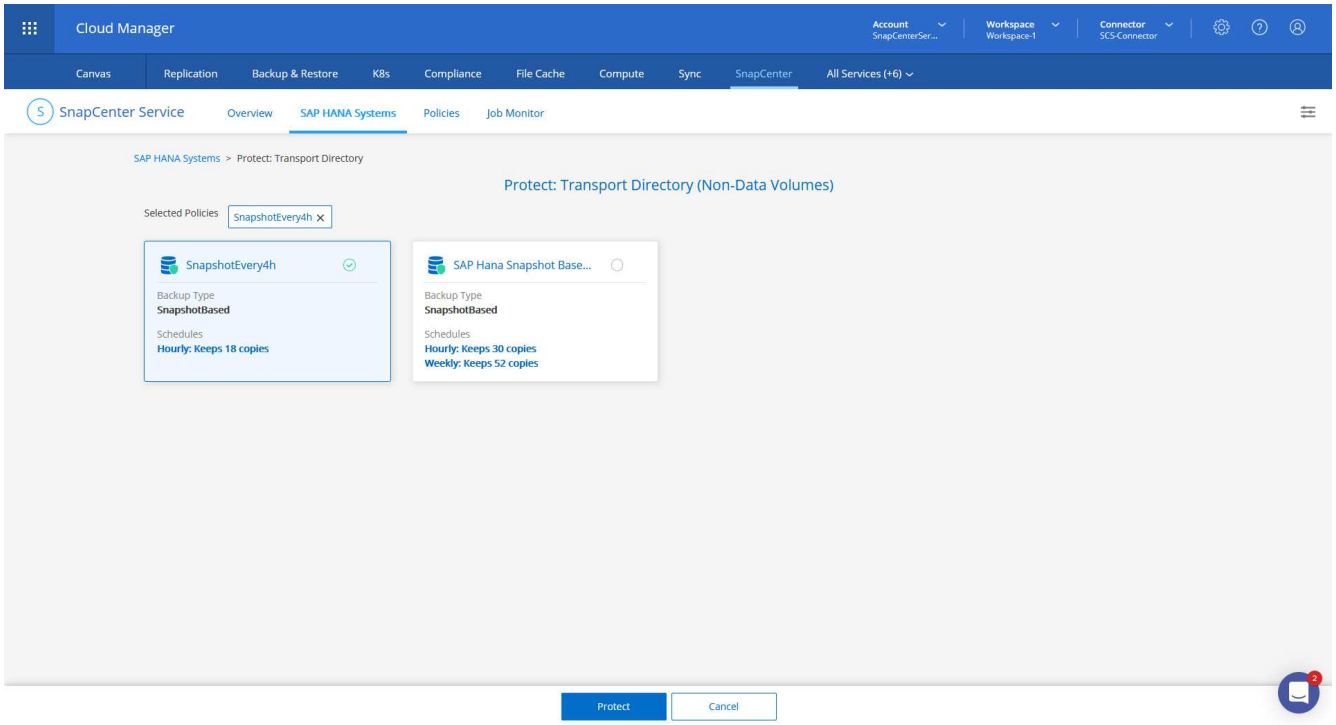
5. Click Add.



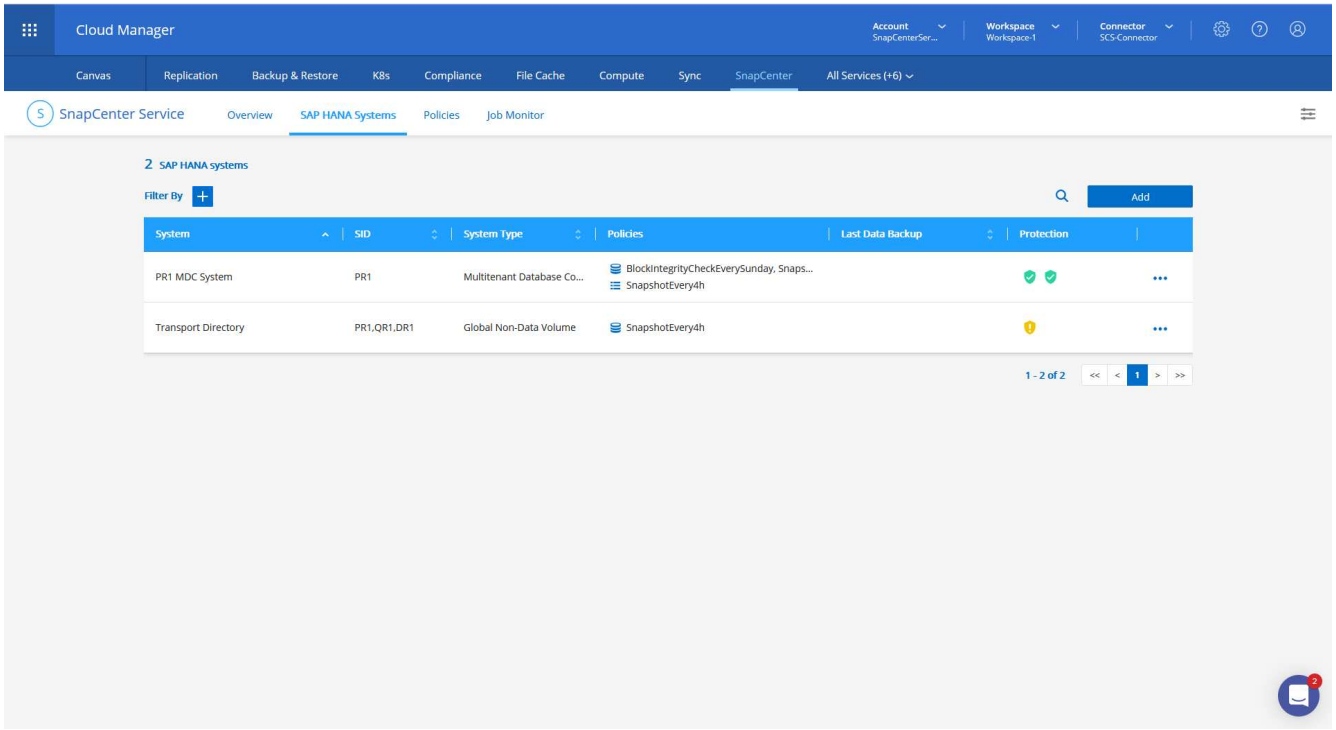
6. To configure the protection, click Proceed.



7. Select the policy and click Protect.



8. Transport Directory now displays in the list of HANA systems as a global non- data volume.



Next: SAP HANA database Snapshot backups.

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